

System of linear Equations

Recall that in two dimensions a line in a rectangular xy-coordinate system can be represented by an equation of the form

$$ax + by = c \quad (a, b \text{ not both } 0)$$

and in three dimensions a plane in a rectangular xyz-coordinate system can be represented by an equation of the form

$$ax + by + cz = d \quad (a, b, c \text{ not all } 0)$$

These are examples of “**linear equations**,” the first being a linear equation in the variables x and y and the second a linear equation in the variables x, y, and z.

A finite set of linear equations is called **a system of linear equations** or, more briefly, a linear system. The variables are called unknowns.

Following are the examples of system of linear equations.

1.
$$\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned}$$

2.
$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}$$

3.
$$\begin{aligned} 3x + 2y &= -5 \\ x - 4y &= 12 \end{aligned}$$

System of linear equations can be written in matrix equation form:

$$\mathbf{A} \mathbf{X} = \mathbf{b} \quad \text{-----} \quad (\mathbf{A})$$

Where $\mathbf{A}$ is the coefficient matrix, $\mathbf{X}$ is the unknown column matrix, and $\mathbf{b}$ is constant column matrix.

Above system of equations defined in previous slide by 2. can be written in matrix eq. form as

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \quad \text{-----} \quad (1)$$

if we write the given system of linear equations in the form

$$\mathbf{AX} = \mathbf{0}$$

Then the system of equations is said to be **Homogeneous**.

Similarly, if we write the given system of equations in the form

$$\mathbf{AX} = \mathbf{b}$$

Then the system of equations is said to be **Non-Homogeneous**.

A system of equations that has no solution is said to be **inconsistent**.

on the other hand, if there exists at least one solution of the system, the system is called **consistent system**.

System of equations can be solved by using following methods:

1. **GAUSS'S Elimination Method**
2. **GAUSS'S Jordan Method**

Before, solve the system of linear equations, we will define **Augmented matrix**.

following matrix equation is the equation of system of linear equations.

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

The matrix obtained by augmenting the matrix A by column vector b is known as “Augmented Matrix”.

Augmented matrix is denoted by A_b .

$$A_b = \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{bmatrix}$$

Solution of System of linear Equations

Gauss's Elimination Method

One of the several methods employed to solve the system of **m** linear equations in **n** variables, is known as Gaussian Elimination method named after its inventor, the famous German mathematician Carl Friedrich Gauss.

Steps of Gaussian Elimination method:

Step#1 Change the system of linear equations to the form $Ax = b$



Step#2 Form the Augmented coefficient matrix A_b by including the elements of b as an extra column in the matrix A



Step#3 Convert the augmented matrix into echelon form by using elementary row operations



Step#4 Find x by detaching the fourth column back to its original position on the right hand side of the matrix equation $AX = b$. this procedure is known as “Backward Substitution”

Example: Use Gaussian Elimination method to solve the system of linear equations ($m = n$)

$$x_1 + 5x_2 + 2x_3 = 9$$

$$x_1 + x_2 + 7x_3 = 6$$

$$-3x_2 + 4x_3 = -2$$

Solution:

Step#1 Change the system of linear equations to the matrix equation form $Ax = b$

$$\begin{bmatrix} 1 & 5 & 2 \\ 1 & 1 & 7 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ -2 \end{bmatrix}$$

Step#2 Form the augmented coefficient matrix.

$$A_b = \begin{bmatrix} 1 & 5 & 2 & 9 \\ 1 & 1 & 7 & 6 \\ 0 & -3 & 4 & -2 \end{bmatrix}$$

Step#3 Convert this augmented coefficient matrix into an echelon form using elementary row operations.

$$\begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & -4 & 5 & -3 \\ 0 & -3 & 4 & -2 \end{bmatrix} \quad R_2 + (-)R_1$$

$$\begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & 1 & \frac{-5}{4} & \frac{3}{4} \\ 0 & -3 & 4 & -2 \end{bmatrix} \quad -\frac{1}{4}R_2$$

$$\begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & 1 & \frac{-5}{4} & \frac{3}{4} \\ 0 & 0 & \frac{1}{4} & \frac{1}{4} \end{bmatrix} R_3 + 3R_2 \qquad \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & 1 & \frac{-5}{4} & \frac{3}{4} \\ 0 & 0 & 1 & 1 \end{bmatrix} 4R_3$$

Step#4 Find x by detaching the fourth column back to its original position on the right hand side of the matrix eq.

$$Ax = b$$

$$\begin{bmatrix} 1 & 5 & 2 \\ 0 & 1 & \frac{-5}{4} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ \frac{3}{4} \\ 1 \end{bmatrix}$$

$$x_1 + 5x_2 + 2x_3 = 9$$

$$x_2 - \frac{5}{4}x_3 = \frac{3}{4}$$

$$x_3 = 1$$

Put $x_3 = 1$ in second last equation $x_2 = 2$ again put these both value in first eq. $x_1 = -3$

Solution of System of linear Equations

by

Gauss- Jordan Method

The Gauss elimination method is based on :

Changing the augmented matrix into echelon form and making the backward substitution.

In Gauss Jordan method, one reduces the augmented matrix into “Reduced Echelon ” form where one does not need the process of backward substitution. This save a lot of manual or computing time.

Steps of Gauss Jordan method:

Step#1 Change the system of linear equations to the form $Ax = b$



Step#2 Form the Augmented coefficient matrix A_b by including the elements of b as an extra column in the matrix A



Step#3 Convert the augmented matrix into Reduced echelon form by using elementary row operations



Step#4 Find x by detaching the fourth column back to its original position on the right hand side of the matrix equation $AX = b$.

Example: Use Gaussian Elimination method to solve the system of linear equations ($m = n$)

$$x_1 + 5x_2 + 2x_3 = 9$$

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$$-3x_2 + 4x_3 = -2$$

Solution:

Step#1 Change the system of linear equations to the matrix equation form $Ax = b$

$$\begin{bmatrix} 1 & 5 & 2 \\ 1 & 1 & 7 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ -2 \end{bmatrix}$$

Step#2 Form the augmented coefficient matrix.

$$A_b = \begin{bmatrix} 1 & 5 & 2 & 9 \\ 1 & 1 & 7 & 6 \\ 0 & -3 & 4 & -2 \end{bmatrix}$$

Step#3 Convert this augmented coefficient matrix into reduced echelon form using elementary row operations.

$$\begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & -4 & 5 & -3 \\ 0 & -3 & 4 & -2 \end{bmatrix} \quad R_2 + (-)R_1$$

$$\begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & 1 & \frac{-5}{4} & \frac{3}{4} \\ 0 & -3 & 4 & -2 \end{bmatrix} \quad -\frac{1}{4}R_2$$

$$\begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & 1 & \frac{-5}{4} & \frac{3}{4} \\ 0 & 0 & \frac{1}{4} & \frac{1}{4} \end{bmatrix} \quad R_3 + 3R_2 \quad \Rightarrow \quad \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & 1 & \frac{-5}{4} & \frac{3}{4} \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad 4R_3$$

$$\begin{bmatrix} 1 & 0 & \frac{33}{4} & \frac{21}{4} \\ 0 & 1 & \frac{-5}{4} & \frac{3}{4} \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad R_1 + (-5)R_2 \quad \Rightarrow \quad \begin{bmatrix} 1 & 0 & \frac{33}{4} & \frac{21}{4} \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad R_2 + \frac{5}{4}R_3$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad R_1 + \left(-\frac{33}{4}\right)R_3$$

Step#4 Find x by detaching the fourth column back to its original position on the right hand side of the matrix eq. $Ax = b$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$$

Expanding from top, we get $x_1 + 0.x_2 + 0.x_3 = -3$

$$0.x_1 + x_2 + 0.x_3 = 2$$

$$0.x_1 + 0.x_2 + x_3 = 1$$

from eq. (i), (ii) and (iii), we have

$$x_1 = -3, \quad x_2 = 2, \quad x_3 = 1$$